



Support Vector Machine

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Analysis

Moons dataset

1. Inferences about the Linear Kernel's performance.

The Linear kernel struggles with the Moons dataset because the data is non-linearly separable. Its decision boundary is a straight line that cuts across both moon-shaped clusters, leading to many misclassifications near the overlap region. Compared to other kernels, this results in lower accuracy and F1-scores, indicating underfitting - the linear model cannot capture the curved structure of the data.

2. Comparison between RBF and Polynomial kernel decision boundaries.

The RBF kernel captures the shape of the data more naturally than the Polynomial kernel. Its boundary smoothly follows the moon curves and adapts well to local variations. The Polynomial kernel, though better than linear, shows slightly irregular and rigid boundaries - it overfits small regions and generalizes less effectively. Hence, RBF provides the best fit and separation for this nonlinear dataset.

Banknote Authentication dataset

1. Which kernel was most effective for this dataset?

The RBF kernel is the most effective for the Banknote Authentication dataset. It gives the highest accuracy and balanced precision-recall scores, successfully capturing the small non-linear variations in the data.

2. Why might the Polynomial kernel have underperformed here?

The Banknote data is mostly linear, so using a Polynomial kernel introduces unnecessary nonlinearity and higher-dimensional transformations. This leads to overfitting, reducing the model's ability to generalize well. The Moons dataset is inherently nonlinear, where a Polynomial kernel is more useful, explaining why its relative performance drops in the Banknote data.

Margin Analysis

1. Which margin (soft or hard) is wider?

The Soft Margin SVM produces a wider margin. A smaller C places less penalty on misclassification, so the optimizer prioritizes maximizing margin width even if some points fall inside or on the wrong side of the boundary.

2. Why does the soft margin model allow "mistakes"?

The Soft Margin SVM allows a few classification errors to create a larger and more flexible margin that generalizes better. It aims to not perfect accuracy on training data but to maximize the margin while minimizing misclassification errors, balancing bias and variance for unseen data.

3. Which model is more likely to be overfitting and why?

The Hard Margin model ($C=100$) is more likely to overfit. A large C forces the SVM to classify training points correctly and heavily penalizes slack, which can lead to a narrow, complex decision boundary that fits noise and outliers instead of the underlying pattern.

4. Which model would you trust more for new data and why?

In unseen data, it is better to believe the Soft Margin model more because it prioritizes generalization. In noisy real-world scenarios, a lower C value is preferred because it creates a softer margin, tolerating small errors and improving robustness against noise.

Screenshots

Training Results

Moons dataset

SVM with LINEAR Kernel <PES2UG23CS365>

	precision	recall	f1-score	support
0	0.85	0.89	0.87	75
1	0.89	0.84	0.86	75
accuracy			0.87	150
macro avg	0.87	0.87	0.87	150
weighted avg	0.87	0.87	0.87	150

SVM with RBF Kernel <PES2UG23CS365>

	precision	recall	f1-score	support
0	0.95	1.00	0.97	75
1	1.00	0.95	0.97	75
accuracy			0.97	150
macro avg	0.97	0.97	0.97	150
weighted avg	0.97	0.97	0.97	150

SVM with POLY Kernel <PES2UG23CS365>

	precision	recall	f1-score	support
0	0.85	0.95	0.89	75
1	0.94	0.83	0.88	75
accuracy			0.89	150
macro avg	0.89	0.89	0.89	150
weighted avg	0.89	0.89	0.89	150

Banknote Authentication dataset

SVM with LINEAR Kernel <PES2UG23CS365>

	precision	recall	f1-score	support
Forged	0.90	0.88	0.89	229
Genuine	0.86	0.88	0.87	183
accuracy			0.88	412
macro avg	0.88	0.88	0.88	412
weighted avg	0.88	0.88	0.88	412

SVM with RBF Kernel <PES2UG23CS365>

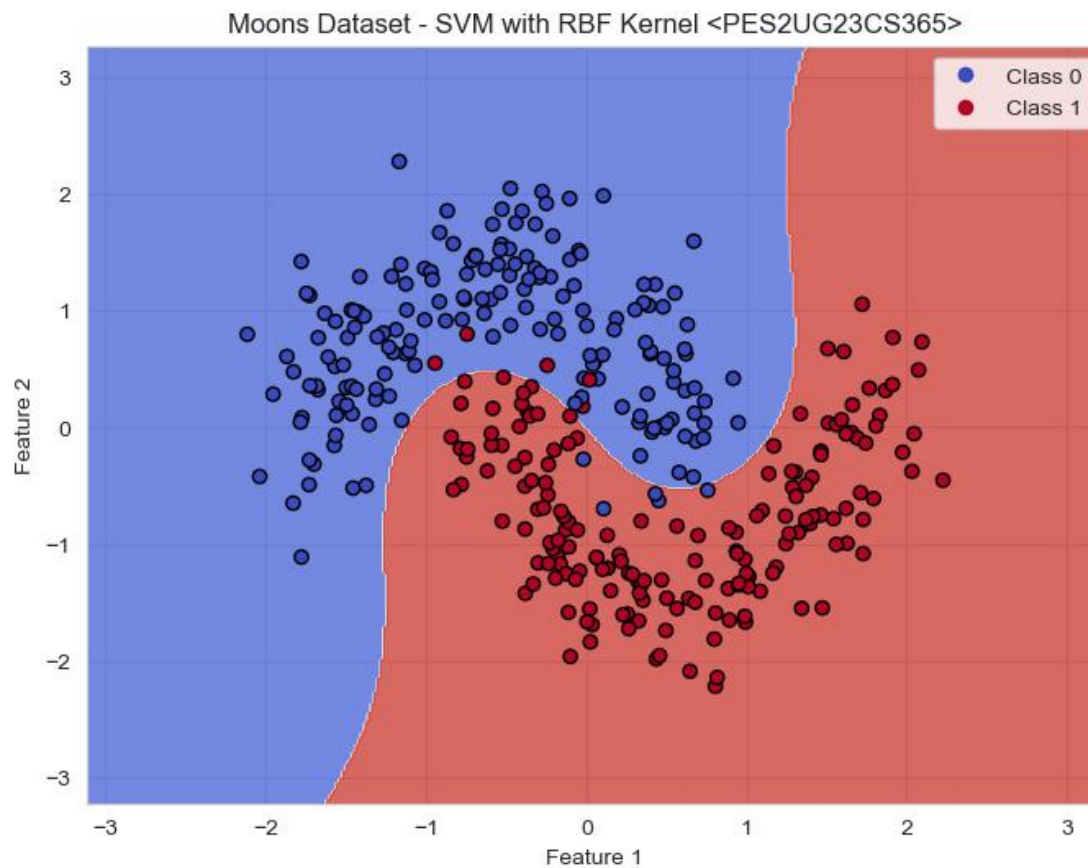
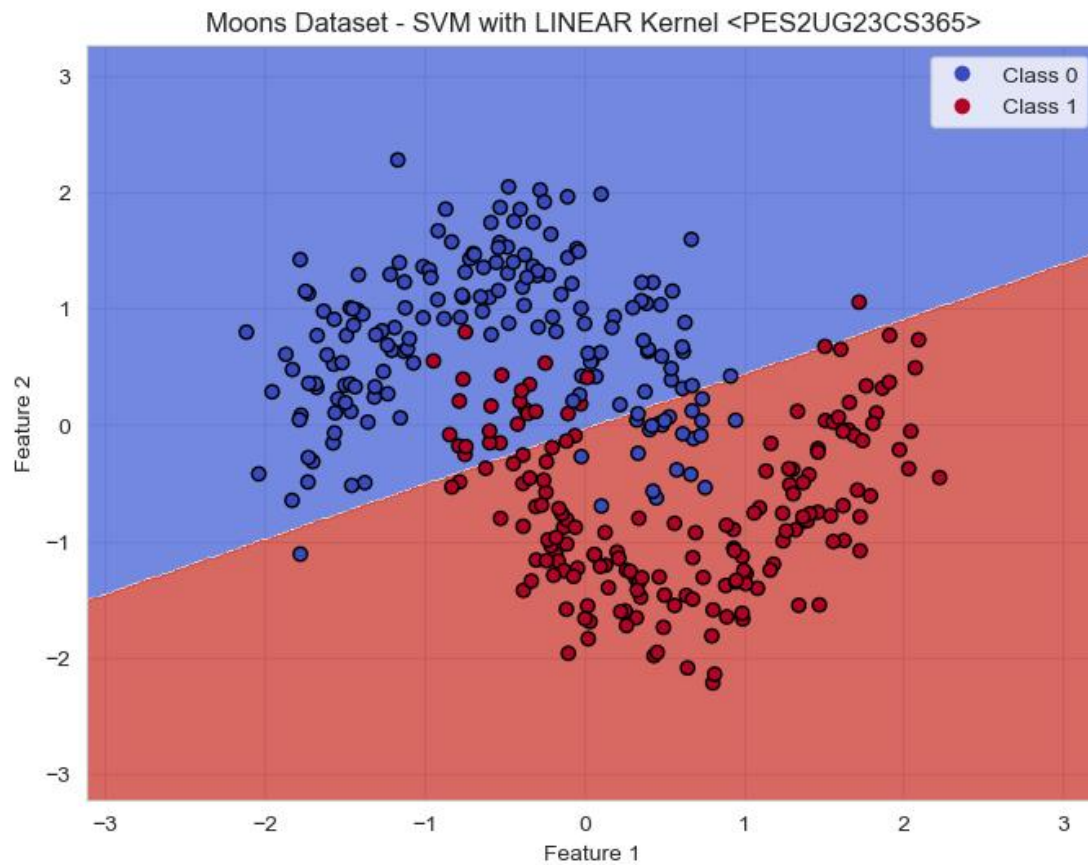
	precision	recall	f1-score	support
Forged	0.96	0.91	0.94	229
Genuine	0.90	0.96	0.93	183
accuracy			0.93	412
macro avg	0.93	0.93	0.93	412
weighted avg	0.93	0.93	0.93	412

SVM with POLY Kernel <PES2UG23CS365>

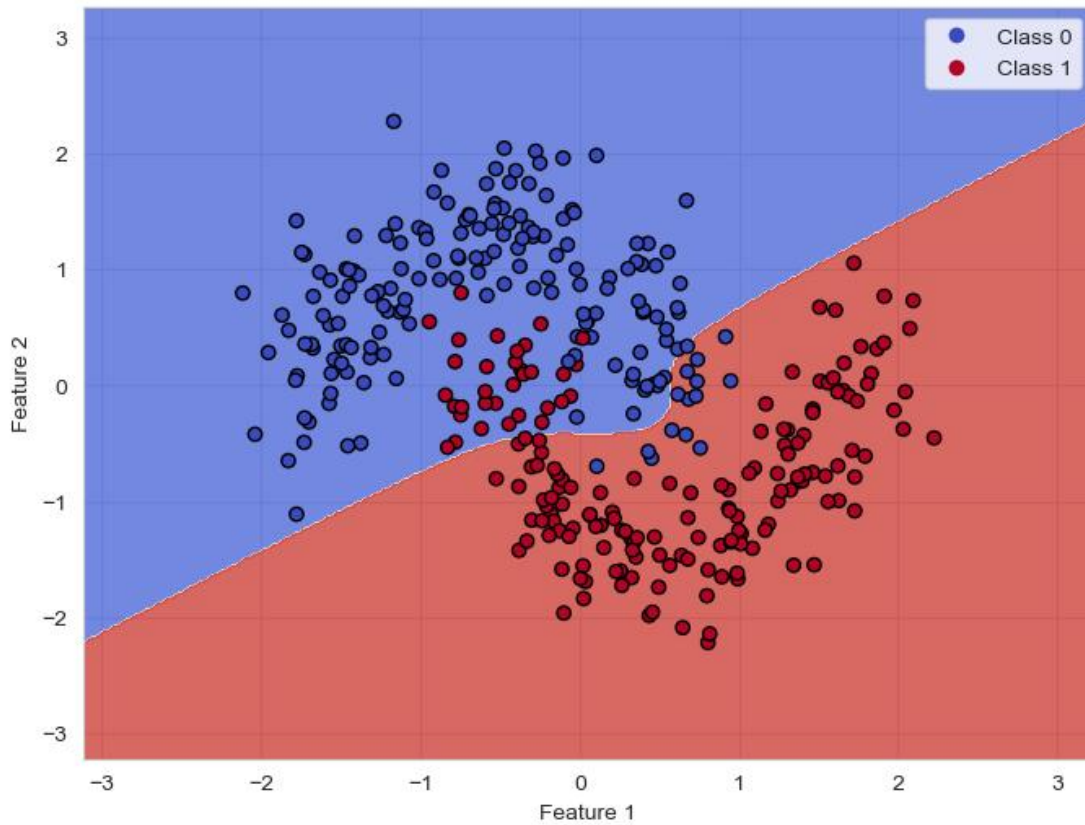
	precision	recall	f1-score	support
Forged	0.82	0.91	0.87	229
Genuine	0.87	0.75	0.81	183
accuracy			0.84	412
macro avg	0.85	0.83	0.84	412
weighted avg	0.85	0.84	0.84	412

Decision Boundary Visuals

Moons dataset

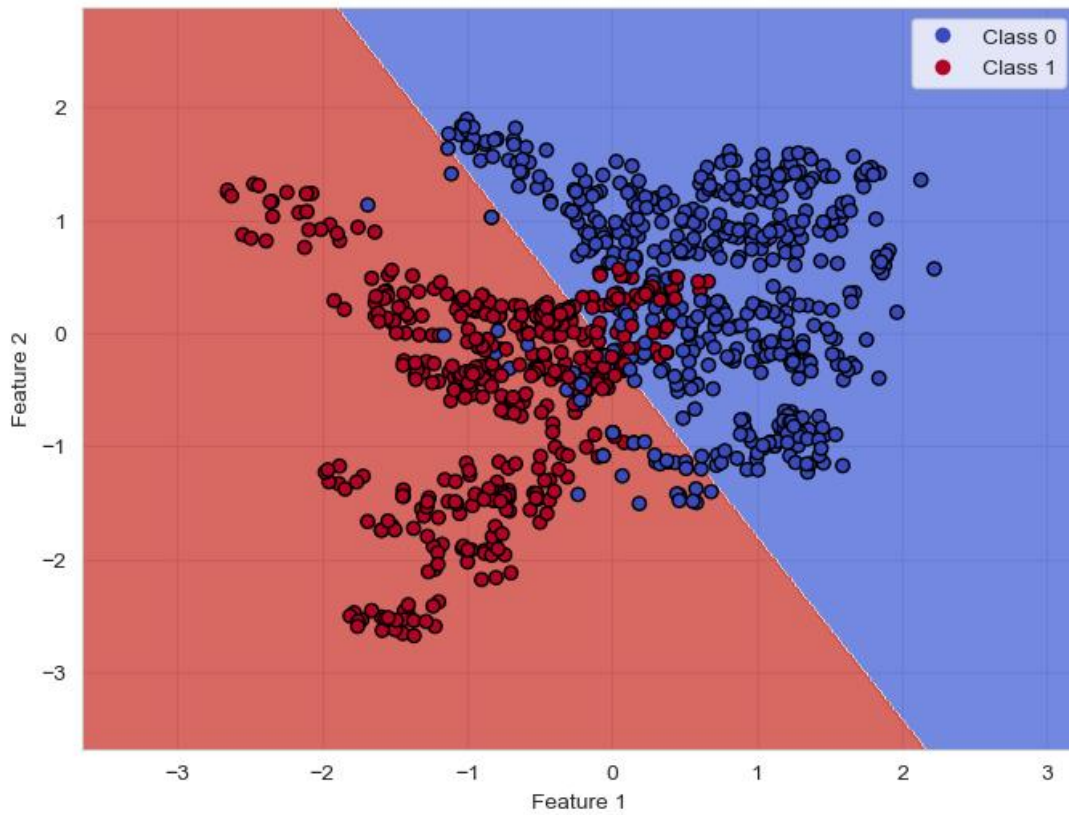


Moons Dataset - SVM with POLY Kernel <PES2UG23CS365>

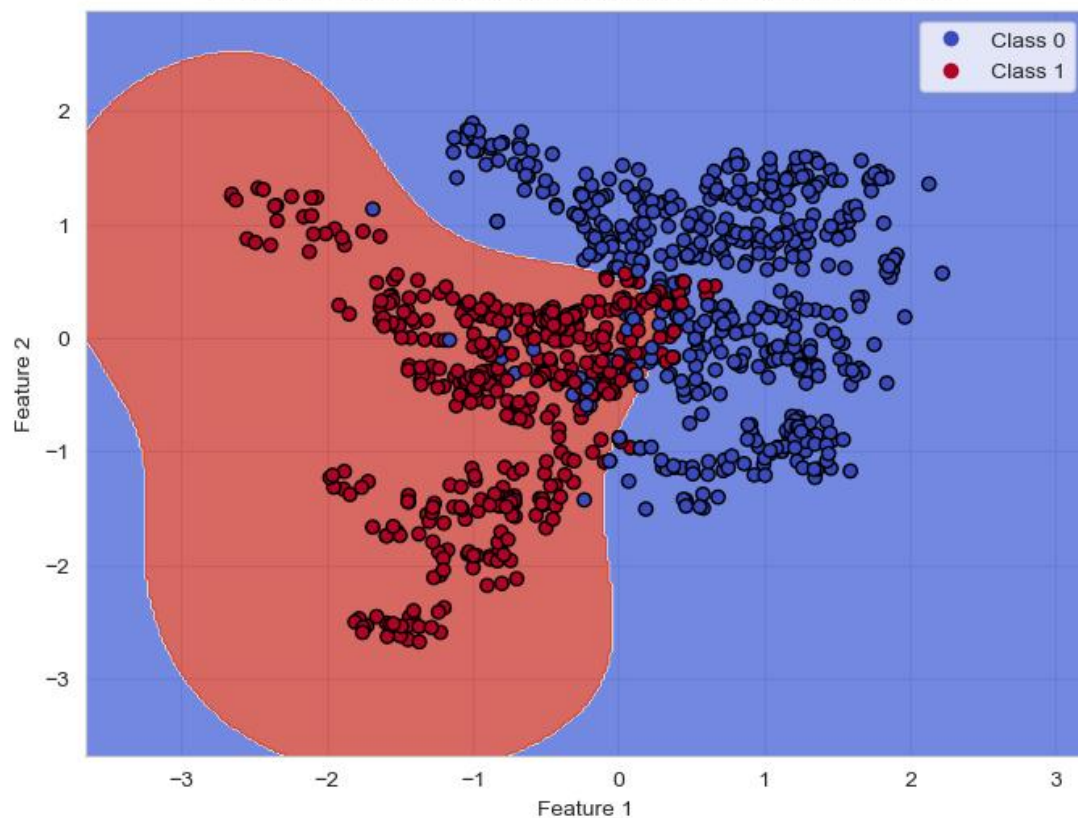


Banknote Authentication dataset

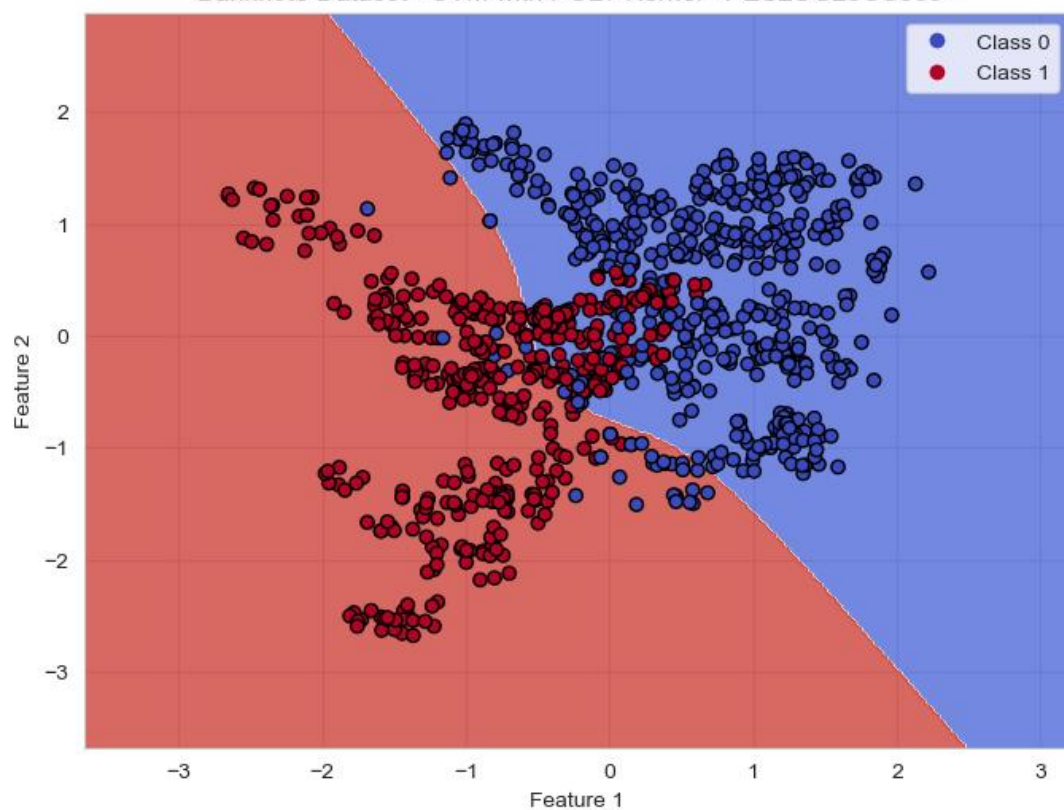
Banknote Dataset - SVM with LINEAR Kernel <PES2UG23CS365>



Banknote Dataset - SVM with RBF Kernel <PES2UG23CS365>

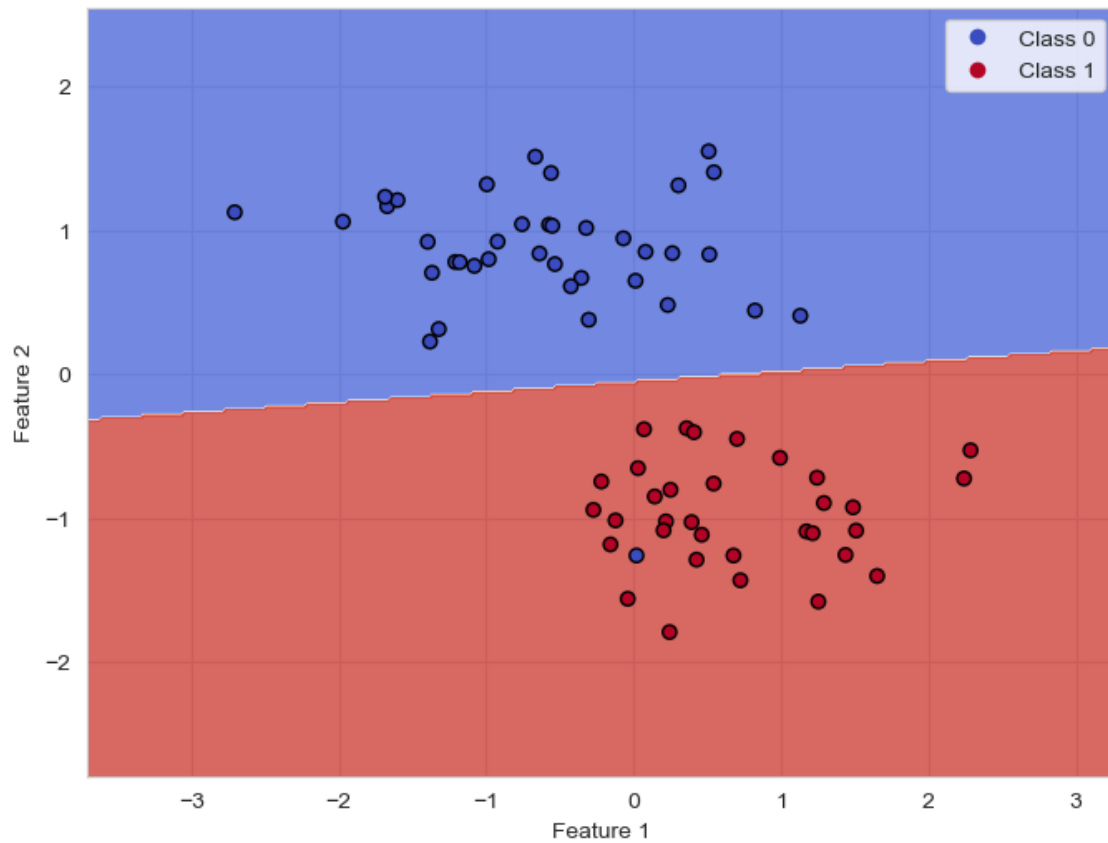


Banknote Dataset - SVM with POLY Kernel <PES2UG23CS365>



Margin Analysis

Hard Margin SVM (C=100) <PES2UG23CS365>



Soft Margin SVM (C=0.1) <PES2UG23CS365>

